

MOHAMED BEN HASSEN

AI ENGINEER | FULL-STACK DEVELOPER

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PROFESSIONAL SUMMARY

AI Engineer and Developer with hands-on experience in computer vision, deep learning, recommender systems, LLM-assisted workflows, MLOps, cloud deployment, modern web development, and system design. Currently developing a custom company-wide digital platform for iPrint International, combining role-based access, workflow automation, document generation, advanced tracking, client portals, and AI-assisted email workflows. Strong focus on building practical systems that connect AI, automation, and product execution.

TECHNICAL SKILLS

Programming: Python, C++, JavaScript, SQL, R, HTML, CSS

AI / ML: TensorFlow, Scikit-Learn, CNNs, YOLOv8, OpenCV, OpenPose, recommender systems, LLM workflows, TF-IDF, SVD, Pandas, NumPy, Matplotlib

Cloud / MLOps / Backend: Azure, Docker, Kubernetes/AKS, MLflow, Airflow, Neptune, Flask, FastAPI, Node.js, PostgreSQL, Cosmos DB

Frontend / Product: React, Vite, Framer Motion, responsive UI/UX, Git/GitHub, Vercel, GitHub Pages

IT / Operations: System Administration, Active Directory, Microsoft 365, automation, security improvement, technical documentation

PROFESSIONAL EXPERIENCE

iPrint International - AI Engineer & Developer (Project-Based)

Doha, Qatar | Aug 15, 2026 - Present

- Developing a custom company-wide digital platform to replace generic ERP dependency and centralize operations across Client Servicing, Finance, Traffic, Production, and other departments.
- Designing a role-based access system where each department sees, updates, and tracks only the workflows, documents, and lifecycle stages relevant to its responsibilities.
- Building automation features for department workflows, including structured document generation, Excel sheet generation, job brief mapping, and reusable data entry flows to reduce manual copy-paste and operational delays.
- Developing an advanced tracking system that allows internal teams and clients to monitor job progress, department-specific status updates, pending actions, and the full production lifecycle through controlled portals.
- Planning and integrating AI-assisted email workflows for Client Servicing, including internal/external email classification, read/unread visibility, request summarization, bullet-point structuring, and extraction of key order details for later mapping into documents and system entries.

VisioRide - Technology Department

Qatar | Sep 9, 2026 - Present

- Supporting technology initiatives across software development, AI integration, automation, and security improvement.
- Contributing to technical planning and implementation for future platform enhancements, internal tools, and intelligent digital features.

Qatar Mobility Innovations Center (QMIC) - AI Engineer Intern

Doha, Qatar | Jun 2024 - Aug 2024

- Vessel detection/classification from satellite imagery: researched and curated relevant datasets, converted annotations to YOLO format with Roboflow, trained and fine-tuned YOLOv8 models, and resolved framework/version issues.
- Self-driving perception optimization: integrated segmentation-based weights and GrabCut-generated masks into the detection workflow, refined data/annotations, and improved model performance by more than 4%.
- Performed model testing, output analysis, and statistical evaluation to compare iterations and validate improvements.

Qatar National Datathon - Data Analyst / Participant

Feb 2025

- Collaborated on an OpenAI-powered analytics website with voice recognition and data visualizations, allowing users to interact conversationally with NPC datasets and explore labor-force insights.
- Built a predictive model to assess STEM-enrollment growth toward Qatar Vision 2030 targets and contributed data-driven recommendations for attracting high-skilled talent and expanding technology-sector opportunities.

EDUCATION

Bahcesehir University | Istanbul, Turkiye

Bachelor of Science - Artificial Intelligence Engineering

Master's Degree - Artificial Intelligence Engineering | GPA: 3.17 / 4.00

SELECTED CERTIFICATIONS

Microsoft Azure Cloud Services • Applied Machine Learning • Machine Learning with Python • DevOps, DataOps & MLOps • MLOps Tools: MLflow & Hugging Face • Python Essentials for MLOps • Neural Networks and Deep Learning • Improving Deep Neural Networks • 9 Kaggle micro-certificates.

LANGUAGES

Arabic - Native | English - C2 / Proficient | French - B2 / Upper Intermediate | Turkish - B1 / Intermediate

TECHNICAL PROJECTS

Deep Learning Systems: CNN Robustness & Failure Modes | *Python, CNNs, Fashion-MNIST*

- Built a fixed CNN and ran controlled one-variable-at-a-time experiments on activation functions, dataset size, and learning rate to study representation, generalization, and optimization failure modes.
- Compared ReLU vs. tanh (~91.5% vs. ~91.8% test accuracy); tested 100%/25%/10%/5% training data; best tested learning rate was 1e-3 with 91.83% test accuracy.

OpenPose Gait Analysis & Joint-Angle Measurement | *OpenPose, OpenCV, Vector Mathematics*

- Used pretrained CMU OpenPose to extract lower-body keypoints from walking sequences and vector dot-products to calculate joint angles for gait analysis and abnormal-pattern monitoring.

Computer Vision & Image Processing | *OpenCV, Homography, Gaussian/Laplacian Pyramids*

- AR billboard compositing: built a replacement pipeline using HSV/morphology and contours for region detection, four-corner ordering, gamma-based brightness matching, perspective homography, and alpha blending; batch-processed background/poster combinations.
- Multi-scale image enhancement: implemented 5-level Gaussian/Laplacian pyramids with weighted reconstruction; processed 13 images with level weights [1.6, 1.4, 1.2, 1.1, 1.0].

News Recommender System | *MIND-small, TF-IDF, SVD, ML, Deep Learning, LLMs*

- Designed and evaluated content-based, collaborative, and hybrid recommenders on Microsoft MIND-small; built TF-IDF user profiles and an implicit user-item matrix with Truncated SVD.
- Experimented with machine-learning, deep-learning-integrated, and LLM-assisted recommendation approaches; analyzed personalization, cold-start, sparsity, and hybrid ranking trade-offs.
- Content-based filtering performed best overall: HR@10 0.673, NDCG@10 0.357, Precision@10 0.078.

Gold Bakery - Luxury E-commerce Frontend | *React, Vite, Framer Motion, Vercel*

- Designed and developed a live responsive bakery storefront with reusable React components, product/order flows, smooth scrolling, advanced motion, and award-inspired transitions. Backend/payment integration is in progress.
- Live: gold-bakery.vercel.app/

Personal Portfolio Website | *React, Framer Motion, GitHub Pages*

- Built and deployed an interactive portfolio showcasing AI, computer vision, cloud/MLOps, and software-engineering projects with responsive layouts and animated sections.
- Live: mohamedbenhasse.github.io/mohamed-portfolio/

Azure Cloud Web Application | *Flask, Docker, AKS, Cosmos DB*

- Built and deployed a containerized Flask app on Azure using Docker, Cosmos DB, and AKS for scalable deployment and orchestration.

ML Lifecycle Management System | *MLflow, TensorFlow, Scikit-Learn*

- Developed an ML lifecycle workflow for experiment tracking, model management, deployment, monitoring, and documentation with MLflow, TensorFlow, and Scikit-Learn.

CS188 Pacman Search & Adversarial AI | *Python, DFS, BFS, A*, Minimax, Alpha-Beta, Expectimax*

- Implemented DFS, BFS, A*, Minimax, Alpha-Beta Pruning, and Expectimax in the Pacman environment, focusing on decision quality and search efficiency.

SELECTED KAGGLE PROJECTS

Titanic: Machine Learning from Disaster | *Python, Pandas, Scikit-Learn, Feature Engineering*

- Built and optimized an end-to-end survival classification workflow covering EDA, missing-value handling, title extraction, family-size features, preprocessing pipelines, and cross-validated model comparison.
- Prepared a clean Kaggle notebook and submission workflow, then published the optimized project to GitHub as part of a professional machine-learning portfolio.
- Live: github.com/mohamedbenhasse/titanic-project-

Tabular Stroke Prediction | *Python, Pandas, Scikit-Learn, Classification*

- Developed and evaluated binary classification models on structured health data, covering data cleaning, categorical/numerical preprocessing, feature engineering, model training, and comparison.
- Focused on practical model evaluation, including accuracy and classification trade-offs, to better understand performance limits on tabular prediction tasks.